

Homework 2, Math 609

September 10, 2026

PROBLEM 1:

- (10 points) Prove that for any matrix $A \in \mathbb{C}^{n \times n}$

$$n^{-1}\|A\|_2 \leq n^{-1/2}\|A\|_\infty \leq \|A\|_2 \leq n^{1/2}\|A\|_1 \leq n\|A\|_2.$$

- (10 points) A weighted ℓ_∞ norm is a norm on $\mathbb{R}^n$ of the form

$$\|x\| = \max_{i=1,\dots,n} w_i |x_i|,$$

where $w_1, \dots, w_n$ are fixed positive numbers called weights. Prove the norm postulates for this norm. What is the subordinate matrix norm?

PROBLEM 3: (10 points) Prove that the $\|\cdot\|_1$ matrix norm can be computed by

$$\|A\|_1 := \max_{x \neq 0} \frac{\|Ax\|_1}{\|x\|_1} = \max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}|$$

PROBLEM 4:

- (5 points) Show that if a square matrix A satisfies an inequality $\|Ax\| \geq \theta\|x\|$ for all x with $\theta > 0$, then A is nonsingular and $\|A\|^{-1} \leq \theta^{-1}$. This is valid for any vector norm and its subordinate matrix norm.
- (5 points) Show that if there is a polynomial P without constant term such that

$$\|I - P(A)\| < 1,$$

then A is invertible. Find a formula for A^{-1} .

PROBLEM 5: Show that if A is a symmetric positive definite matrix then

- (5 points) $a_{ii} > 0$ for each $i = 1, 2, \dots$
- (5 points) $a_{ij}^2 < a_{ii}a_{jj}$ for each $i \neq j$ (Hint: Consider a vector $\mathbf{x}$ with $\alpha \in \mathbb{R}$ on position i , 1 on position j and zeroes everywhere else and analyse $\mathbf{x}^T A \mathbf{x}$).